



BOARD QUESTION PAPER : OCTOBER 2015

PHYSICS

Time: 3 Hours**Total Marks: 70****Note:**

- i. All questions are compulsory.
- ii. Neat diagrams must be drawn wherever necessary.
- iii. Figures to the right indicate full marks.
- iv. Use of only logarithmic table is allowed.
- v. All symbols have their usual meaning unless otherwise stated.

SECTION – I

- Q.1. A.** State an expression for the moment of inertia of a solid uniform disc, rotating about an axis passing through its centre, perpendicular to its plane. Hence derive an expression for the moment of inertia and radius of gyration:
- i. about a tangent in the plane of the disc, and
 - ii. about a tangent perpendicular to the plane of the disc.
- B.** In a set, 21 tuning forks are arranged in a series of decreasing frequencies. Each tuning fork produces 4 beats per second with the preceding fork. If the first fork is an octave of the last fork, find the frequencies of the first and tenth fork. [7]

OR

- A.** Discuss the composition of two S.H.M.s along the same path having same period. Find the resultant amplitude and initial phase.
- B.** A sonometer wire is in unison with a tuning fork of frequency 125 Hz when it is stretched by a weight. When the weight is completely immersed in water, 8 beats are heard per second. Find the specific gravity of the material of the weight. [7]
- Q.2. Select and write the most appropriate answer from the given alternatives for each sub-question :** [7]

- i. Which of the following substances is ductile?

(A) Glass	(B) High carbon steel
(C) Steel	(D) Copper
- ii. Angle of contact for the pair of pure water with clean glass is _____.

(A) acute	(B) obtuse
(C) 90°	(D) 0°
- iii. A seconds pendulum is suspended in an elevator moving with constant speed in downward direction. The periodic time (T) of that pendulum is _____.

(A) less than two seconds	(B) equal to two seconds
(C) greater than two seconds	(D) very much greater than two seconds
- iv. The equation of a progressive wave is $y = 7 \sin (4t - 0.02x)$, where x and y are in cms and time t in seconds. The maximum velocity of a particle is _____.

(A) 28 cm/s	(B) 32 cm/s
(C) 49 cm/s	(D) 112 cm/s
- v. The dimensions of emissive power are

(A) $[M^1 L^{-2} T^{-3}]$	(B) $[M^1 L^2 T^{-3}]$
(C) $[M^1 L^0 T^{-3}]$	(D) $[M^1 L^0 T^{-2}]$



- vi. The pressure (P) of an ideal gas having volume (V) is $\frac{2E}{3V}$, then the energy E is _____.
(A) translational kinetic (B) rotational kinetic
(C) vibrational kinetic (D) inversely proportional to pressure
- vii. The fundamental frequency of transverse vibration of a stretched string of radius r is proportional to _____.
(A) r^{-2} (B) r^{-1}
(C) $r^{-\frac{1}{2}}$ (D) r^2

Q.3. Attempt any SIX:

[12]

- Draw a neat labelled diagram of conical pendulum. State the expression for its periodic time in terms of length.
- A raindrop of diameter 4 mm is about to fall on the ground. Calculate the pressure inside the raindrop. [Surface tension of water $T = 0.072$ N/m, atmospheric pressure $= 1.013 \times 10^5$ N/m²]
- Discuss the weightlessness experienced by an astronaut in an orbiting satellite.
- The periodic time of a linear harmonic oscillator is 2π second, with maximum displacement of 1 cm. If the particle starts from extreme position, find the displacement of the particle after $\frac{\pi}{3}$ seconds.
- State and prove : Law of conservation of angular momentum.
- A pinhole is made in a hollow sphere of radius 5 cm whose inner wall is at temperature 727°C . Find the power radiated per unit area. [Stefan's constant $\sigma = 5.7 \times 10^{-8}$ J/m²s K⁴, emissivity (e) = 0.2]
- Draw a neat labelled diagram showing forces acting on the meniscus of water in a capillary tube.
- Compute the temperature at which the r.m.s. speed of nitrogen molecules is 832 m/s. [Universal gas constant, $R = 8320$ J/k mole K, molecular weight of nitrogen = 28.]

Q.4. Attempt any THREE:

[9]

- Discuss the behaviour of wire under increasing load.
- Determine the binding energy of satellite of mass 1000 kg revolving in a circular orbit around the Earth when it is close to the surface of Earth. Hence find kinetic energy and potential energy of the satellite.
[Mass of Earth $= 6 \times 10^{24}$ kg, radius of Earth = 6400 km; gravitational constant $G = 6.67 \times 10^{-11}$ Nm²/kg²]
- Show that all harmonics are present on a stretched string between two rigid supports.
- A stone of mass 100 g attached to a string of length 50 cm is whirled in a vertical circle by giving velocity at lowest point as 7 m/s. Find the velocity at the highest point.
[Acceleration due to gravity $= 9.8$ m/s²]

SECTION – II

- Q.5. A.** Obtain an expression for average power dissipated in a purely resistive A.C. circuit. Define power factor of the circuit and state its value for purely resistive A.C. circuit.
- B.** A rectangular coil of a moving coil galvanometer contains 50 turns each having area 12 cm². It is suspended in radial magnetic field 0.025 Wb/m² by a fibre of twist constant 15×10^{-10} N-m/degree. Calculate the sensitivity of the moving coil galvanometer.

[7]



OR

- A. State Bohr's third postulate for hydrogen (H_2) atom. Derive Bohr's formula for the wave number. Obtain expressions for longest and shortest wavelength of spectral lines in ultraviolet region for hydrogen atom.
- B. The photoelectric current in a photoelectric cell can be reduced to zero by a stopping potential of 1.8 volt. Monochromatic light of wavelength $2200\text{\AA}$ is incident on the cathode. Find the maximum kinetic energy of the photoelectrons in joules. [Charge on electron = $1.6 \times 10^{-19}\text{ C}$]

[7]

Q.6. Select and write the most appropriate answer from the given alternatives for each sub-question :

[7]

- Which one of the following particles cannot be accelerated by a cyclotron?
 (A) Electrons (B) Protons
 (C) Deuterons (D) α - particles
- In biprism experiment two interfering waves are produced due to division of _____.
 (A) amplitude (B) wavefront
 (C) amplitude and wavefront (D) neither wavefront nor amplitude
- The output of NOR gate is high, when _____.
 (A) all inputs are high (B) all inputs are low
 (C) only one of its inputs is high (D) only one of its inputs is low
- Light of a certain wavelength has a wave number $\bar{\nu}$ in vacuum. Its wave number in a medium of refractive index n is _____.
 (A) $\frac{n}{\bar{\nu}}$ (B) $\frac{1}{n\bar{\nu}}$
 (C) $\frac{\bar{\nu}}{n}$ (D) $n\bar{\nu}$
- If the radius of a sphere is doubled without changing the charge on it, then electric flux originating from the sphere is _____.
 (A) double (B) half
 (C) same (D) zero
- The momentum of a photon of de Broglie wavelength $5000\text{\AA}$ is _____.
 [Planck's constant = $6.63 \times 10^{-34}\text{ J.s.}$]
 (A) $1.326 \times 10^{-28}\text{ kg-m/s}$ (B) $7.54 \times 10^{-28}\text{ kg-m/s}$
 (C) $1.326 \times 10^{-27}\text{ kg-m/s}$ (D) $7.54 \times 10^{-27}\text{ kg-m/s}$
- Ionosphere mainly consists of _____.
 (A) positive ions and electrons (B) water vapour and smoke
 (C) ozone layer (D) dust particles

Q.7. Attempt any SIX:

[12]

- State any 'two' possible sources of errors in meter-bridge experiment. How can they be minimised?
- A potentiometer wire has resistance of per unit length of $0.1\ \Omega/\text{m}$. A cell of e.m.f. 1.5V balances against 300 cm length of the wire. Find the current in the potentiometer wire.
- Give any 'two' points of differences between diamagnetic and ferromagnetic substances.
- An iron rod of area of cross-section 0.1m^2 is subjected to a magnetising field of 1000 A/m . Calculate the magnetic permeability of the iron rod.
 [Magnetic susceptibility of iron = 59.9, magnetic permeability of vacuum = $4\pi \times 10^{-7}\text{ S. I. unit}$]
- Draw a neat labelled circuit diagram of experimental arrangement for study of photoelectric effect.



- vi. A coil of 100 turns, each of area 0.02 m^2 is kept in a uniform field of induction $3.5 \times 10^{-5} \text{ T}$. If the coil rotates with a speed of 6000 r.p.m. about an axis in the plane of the coil and perpendicular to the magnetic induction, calculate peak value of e.m.f. induced in the coil.
- vii. Define modulation and transducer.
- viii. In a biprism experiment, when a convex lens was placed between the biprism and eyepiece at a distance of 30 cm from the slit, the virtual images of the slits are found to be separated by 7 mm. If the distance between the slit and biprism is 10 cm and between the biprism and eyepiece is 80cm, find the linear magnification of the image.

Q.8. Attempt any THREE:

[9]

- i. With the help of a neat circuit diagram, explain the working of a photodiode. State its any 'two' uses.
- ii. A parallel beam of monochromatic light is incident on a glass slab at an angle of incidence 60° . Find the ratio of width of the beam in the glass to that in the air if refractive index of glass is $3/2$.
- iii. With the help of neat diagram, explain how non-polar dielectric material is polarised in external electric field of increasing intensity. Define polarisation in dielectrics.
- iv. In a single slit diffraction pattern, the distance between first minima on the right and first minima on the left of central maximum is 4 mm. The screen on which the pattern is displaced, is 2m from the slit and wavelength of light used is $6000 \text{ \AA}$. Calculate width of the slit and width of the central maximum.